A study of wolves and birds populations in the Alaska region

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Introduction

The presence of a single predator can change a lot in an ecosystem and create trophic cascades. This phenomenon is one of the most influential concepts in ecology since the 1980s, as it describes the sometimes drastic changes in the food chain or biodiversity that occur when removing or adding a top predator to an ecosystem (Ripple et al., 2016). The study of bird populations in relation to the presence and evolution of wolf populations has already helped to visualize the mechanics of trophic cascades (Dobson, 2014 ; Hebblewhite et al., 2005). When apex predators like wolves regulate population density and alter the foraging habits of large herbivores, they indirectly relieve the intense browsing pressure on crucial plant life (Hebblewhite et al., 2005 ; Ripple et al., 2001). These events help close vegetation such as willows and aspen regenerate, creating the habitat and resources necessary to support rich diversity and higher abundance of songbirds (Dobson, 2014 ; Hebblewhite et al., 2005).

While the direct impact of top predators on their immediate prey is well-documented, the extended trophic consequences on non-target species are not as clear. Specifically, the evolution and demographic shifts of bird populations in response to changing wolf densities have yet to be quantified. Our core problem lies in determining whether the presence or decline of a keystone predator, here wolves, impacts the abundance of avian species within a shared ecosystem. Regarding the impact of wolves on riparian vegetation (Hebblewhite et al., 2005 ; Ripple et al., 2001), we hypothesize that abundance of bird populations in Mount Kinley in Alaska can be predicted by tracking the evolution of wolf populations, and especially Denali wolves, in the same area. Because Denali wolves act as the primary biological trigger for habitat recovery by relieving herbivore browsing pressure, we anticipate a strong, positive correlation between Denali wolf population stability and overall avian presence.

Problem : *Does the presence of Denali wolves near Mount Kinley in Alaska have a direct positive impact on avian populations ?*

$\mathbf{H_0}$ **:** The presence or evolution of the Denali wolf population has no predictive effect, or has a negative effect on bird population abundance. There is no evidence of a trophic cascade effect.

$\mathbf{H_1}$ **:** The presence or stability of the Denali wolf population has a direct and positive effect on the abundance of bird populations, giving proof for the existence of a trophic cascade through vegetation regeneration.

To test this hypothesis, our methodological framework is grounded in Bayesian statistics. Trophic cascades are inherently noisy, and the indirect pathways connecting apex predators to songbirds carry compounding layers of environmental and observational uncertainty. Rather than generating a single, deterministic forecast, Bayesian inference yields dynamic probability distributions. As continuous field data on wolf metrics and vegetation recovery are introduced, we can iteratively update our models. This framework enables us to explicitly quantify the uncertainty at each node of the food web, providing a mathematically robust, probabilistic assessment of how apex predator dynamics drive avian biodiversity.

Trophic cascades : historical context, modern tools and pratical uses

The scientific formalization of the influence of predators on their environments is relatively recent. Historically, population dynamics were primarily viewed from a "bottom-up" only perspective. Early observations date back to the 1940s with Aldo Leopold, who documented deer overgrazing following the eradication of wolves in North America. However, it was in 1960 that the hypothesis of "top-down" regulation was proposed by Hairston, Smith, and Slobodkin (Hairston, Smith, Slobodkin, 1960). The specific term "trophic cascade" was first used in 1980 by ecologist Robert Paine, initially to describe intertidal marine ecosystems (Paine, 1980). The application of this concept to top predators reached a turning point in 1995 during the reintroduction of wolves to Yellowstone National Park, which provided an opportunity to observe an ecosystem restructure itself and demonstrated how top predators impact ecosystem function and stability (Baker et al., 2016).

To predict ecosystem responses to predator reintroduction without relying on deterministic certainty, scientists utilize methods drawn from Bayesian computation. This allows for ensemble ecosystem modeling, which accounts for the intricate qualitative interaction networks and predicts a probability distribution of plausible ecosystem outcomes (Baker et al., 2016). Today, modeling trophic cascades has practical applications in conservation and environmental policy. It is crucial for assessing the ecosystem-wide implications of interventions such as assisted migration, biocontrol, invasive species eradication, and the "rewilding" of apex predators like wolves or dingoes to restore degraded habitats (Baker et al., 2016).

Data

To conduct our analysis, we use two datasets available at the following addresses : [https://www.nps.gov/dena/learn/nature/wolf-research.htm], [https://doi.org/10.15468/dl.v6w3s3].

The first dataset focuses on Denali wolves and was compiled from long-term field observations conducted in Denali National Park, Alaska. It provides annual estimates of the resident wolf population and its density north of the Alaska Range, based on biannual surveys carried out since 1986 (see **Figure 1**).

The second dataset is an ornithological record sourced from the Global Biodiversity Information Facility (GBIF), specifically extracted from the eBird Observation Dataset. It consists of 46,329 observational records spanning from 1985 to 2024, filtered within a precise geographic polygon corresponding to Denali National Park, Alaska. This dataset registers verified coordinates of bird occurrences and individual counts, providing a robust timeline of local avian presence to pair with the wolf tracking data (see **Figure 1**).

We first loaded the datasets and handled missing values by setting unrecorded bird counts to a baseline of one individual. Next, we aggregated the ornithological data by year to calculate both the total sightings and cumulative individuals. Because the annual number of observations varies significantly and does not necessarily reflect true bird abundance, we computed a ratio of total individuals to total sightings to obtain a normalized metric, before merging this summary with the wolf data for final analysis, which will use the ratio metric to estimate bird population evolution with regards to the wolf population (see **Figure 2**).

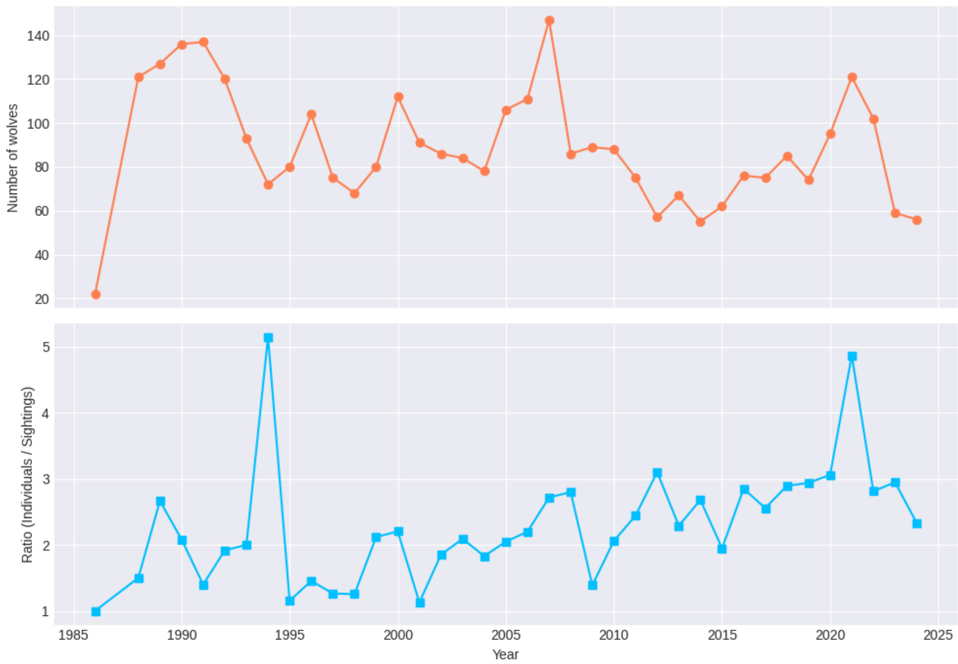

FIGURE 1 – Evolution of the wolf population (graph a) and bird ratio (graph b) over time

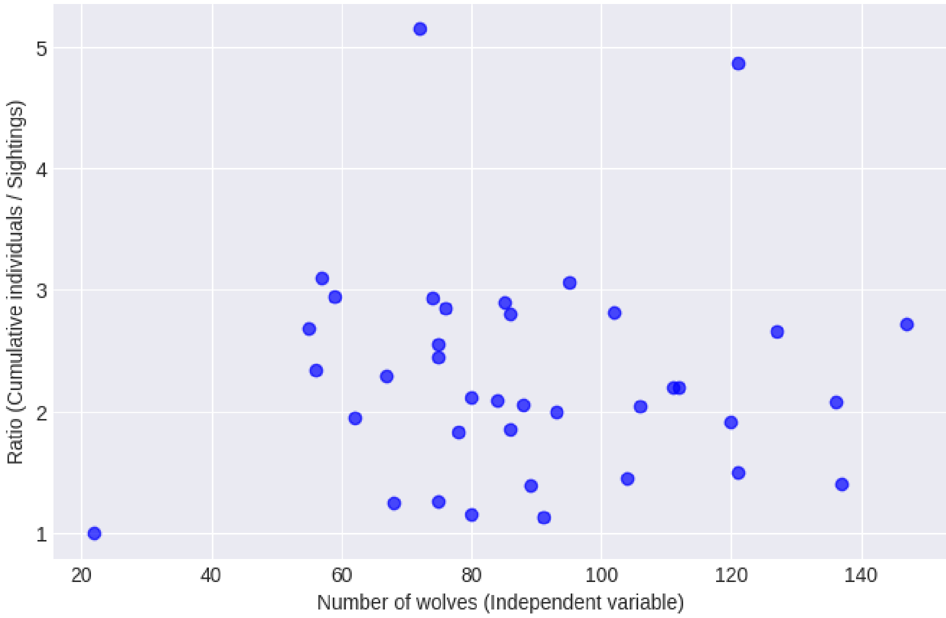

Figure 2 – Bird ratio in relation to the number of wolves

Methods

Helped by the Python package *Arviz*, we conducted our analysis with bayesian tools : bayesian regression, priors adapted, likelihood and Markov Chain Monte Carlo. As a basis of understanding, we defined this tools as such :

— *Bayesian linear regression :*

Historically, the term regression meant "regression to the mean", but nowadays, it is often used to characterize models involving the use of predictor variables to model the distribution of outcome variables. This strategy implies assuming that our predictor variable has a perfectly constant and additive relationship to the mean of the outcome variable ; the model will consider each possible combination of the parameters values to best fit our data. Thus, the posterior distribution ranks these combinations by their plausibility, which provides multiple ones for each possible strength of association, given the model's assumption.

— *Priors adapted :*

A prior probability distribution over statistical hypotheses expresses an uncertain opinion on which of the hypotheses is right. We consider multiple priors over the statistical hypotheses, and compare the performance of these priors on the sample data as if the priors were themselves hypotheses. They define the initial information state of the model before seeing the data.

— *Likelihood :*

The likelihood defines the plausibility of individual observations. By studying the shape of the likelihood distribution, we are able to access precision on the parameters of our model. We can then describe the statistical modeling with acute understanding of its parameters.

The term "likelihood" has been in use in English since at least late Middle English. Its formal use to refer to a specific function in mathematical statistics was proposed by Ronald Fisher, in two research papers published in 1921 and 1922.

— *Markov Chain Monte Carlo :*

For our analysis, we decided to use a Markov Chain Monte Carlo in order to draw samples from our probability distribution. As our problem concerns population density, we found the MCMC appropriate to analyse the variance of the variables studied.

The development of MCMC method takes roots in the exploration of Monte Carlo techniques mostly in physics in the mid-20th century. It was originally designed to endure heavy integration problems using the primitive computer structures. The ease of implementation of the sampling method helped the MCMC become used in mainstream statistics ; the evolution of this method represents a shift in paradigm of statistical computation, enabling numerous complex models being resolved and further expanding the scope and impact of statistics.

More specifically, the linear regression model we use examines the relationship between wolf population size and bird population abundance. Both variables are continuous, and this model provides a straightforward way to assess whether changes in wolf populations are associated with changes in bird abundance.

To test our hypothesis, we modeled the relationship between wolf abundance and bird relative abundance as :

$$\text{BirdRelativeAbundance}_i \sim \mathcal{N}(\mu_i, \sigma) \tag{1}$$
$$\mu_i = \alpha + \beta \times \text{Wolves}_i \tag{2}$$

$$\alpha \sim \mathcal{N}(0, 100) \tag{3}$$
$$\beta \sim \mathcal{N}(0, 10) \tag{4}$$
$$\sigma \sim \text{HalfNormal}(10) \tag{5}$$

where :

— $\text{BirdRelativeAbundance}_i$ is the observed data at site i,

— α represents the expected baseline bird relative abundance in the hypothetical absence of wolves,

— β quantifies the direction and strength of the predictive association between wolf abundance and bird abundance, and

— σ captures residual ecological and observational variability not explained by the model.

To complete the Bayesian specification, we assigned weakly informative prior distributions to our parameters. This approach allows the field data to dominate the posterior inferences while providing enough mathematical regularization to ensure the MCMC sampler converges efficiently and avoids exploring biologically impossible parameter spaces. Posterior distributions were estimated using MCMC sampling implemented via the *PyMC* library in Python.

Results

Model convergence was assessed and confirmed by ensuring all Gelman-Rubin diagnostic values ($\hat{R}$) were strictly less than 1.01 and that bulk Effective Sample Sizes (ESS) exceeded 400 for all parameters, indicating reliable posterior estimates.

TABLE 1 – Posterior estimates of model parameters (89% credible intervals)

Parameter	**Mean**	**SD**	**89% CI lower**	**89% CI upper**
α (Intercept)	2.18	0.52	1.35	3.01
β (Wolves)	0.001	0.006	-0.008	0.010
σ (Residual SD)	0.93	0.11	0.77	1.12

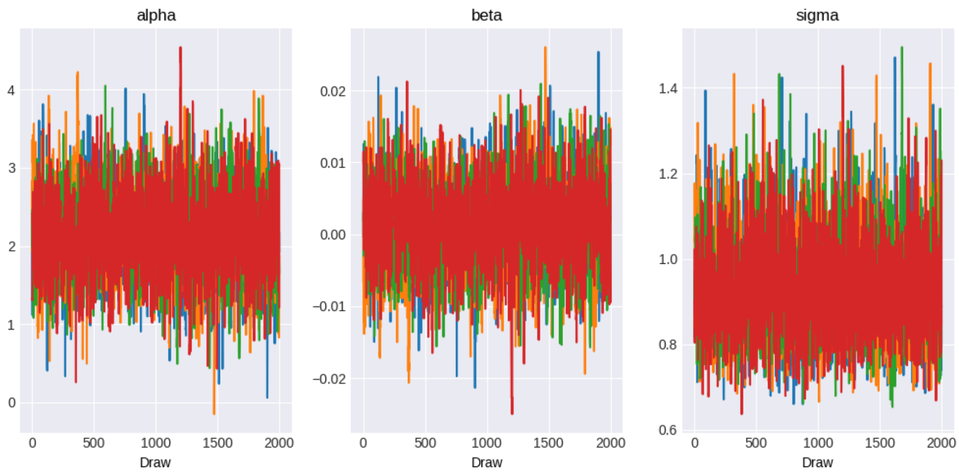

FIGURE 3 – MCMC trace plots to visually check for proper chain mixing

There is no clear evidence of a linear effect of wolf abundance on the bird ratio (see **Figure 4**) : the posterior distribution of the slope parameter β is centered near zero, and its credible interval includes both negative and positive values ($\beta = 0.001$, included in the 89% credible interval, see **Table 1** and **Figure 3**). The residual standard deviation is relatively large compared to the estimated slope, suggesting substantial unexplained variability in the response variable ($\sigma = 0.93$, see **Table 1** and **Figure 3**).

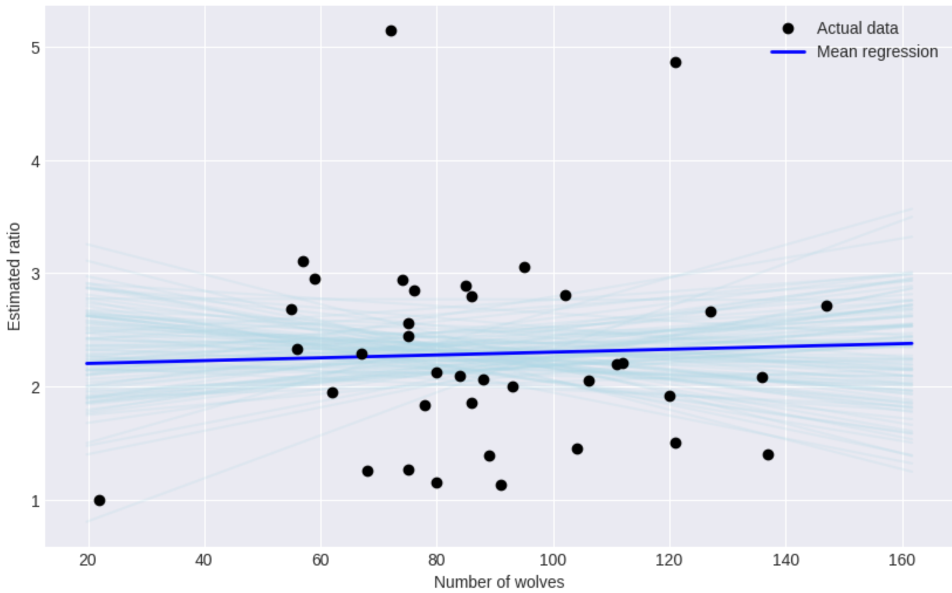

FIGURE 4 – Bayesian Regression : Bird Ratio with regards to Wolf Population

Discussion

Our results do not support a clear relationship between wolf abundance and bird population. The main limitation concerns the modeling. Or linear regression is overly simplistic and includes only one explanatory variable. Ecological systems are complex, and bird populations are likely influenced by many other factors such as habitat structure, food availability, climate conditions, or seasonal effects. The model is not complex enough to capture such subtle ecological interactions. Future work could extend this analysis by including additional ecological covariates or using hierarchical models to account for spatial or temporal structure. Knowing that the effect of wolves on birds is indirect, it could be interesting to test a non linear model.

Conclusion

This project was designed as an exploratory study rather than a definitive ecological analysis. Its main objective was not only to study a potential relationship between wolves and birds, but also to apply and understand Bayesian statistical methods in a concrete research context. As a result, several methodological choices were guided by pedagogical constraints rather than by ecological completeness. We relied on a simple linear model and a limited set of variables, which restricts the type of relationships that can be identified. This simplification was necessary to keep the model interpretable and compatible with our level of statistical expertise, but it also reduces the scope of the conclusions that can be drawn from the results.

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